

Tailoring Ultra-High-Order Optical Skyrmions

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As quasiparticles with topological spin textures, skyrmions have drawn considerable interest from the condensed matter and structured wave communities, offering potential for next-generation high-density, robust information technologies. Nevertheless, generating high-order skyrmions remains a significant challenge, as they have not been reliably observed in any physical system. Here, the method is proposed to create and control ultra-high-order skyrmions (skyrmion number up to 400th) in a structured light system. The topological state transition between bimerons and skyrmions is also experimentally controlled, the transverse size of an arbitrary-order skyrmionic beam is tailored independently of the topological number, and ensure the topological stability upon propagation. This work provides solutions for topologically resilient communication and memory, significantly enhancing information capacity.

1. Introduction

Skyrmions, originally proposed by the British particle physicist Tony Skyrme as topological spin textures to describe localized meson fields, are quasiparticles that may or may not possess an effective mass.^[1] The nontrivial topological structure of skyrmions, which governs both their static properties and dynamic behavior in magnetic domains, has made them a subject of extensive research in condensed matter physics.^[2–5] Magnetic skyrmions, excited by out-of-plane magnetic fields at material surfaces, can be extremely small and can be driven by weak currents. This holds promise for next-generation high-density information storage and

logic manipulation devices.^[6–8] Because topological types and numbers are the primary information carriers for skyrmions, constructing diverse, higher-order quasiparticles is critical. For example, merons with fractional topological charges,^[9] bimerons as homotopy classes of skyrmions,^[10] skyrmion bags,^[11] and skyrmion bundles^[12] reported in recent years. In the fields of nonlinear optics^[13] and condensed matter,^[14] efforts to generate high-order skyrmions have been made. However, for example, magnetic materials tend to favor lower energy states, making the generation of higher-order skyrmions extremely difficult in any physical system.

Optical skyrmions are initially generated by disrupting the transverse nature of electromagnetic fields within evanescent waves, encompass both field and spin skyrmions.^[15–17] The polarization of light, which embodies its wave nature, directly correlates with the spin of photons. Consequently, skyrmions can be created in free space through suitable polarization configurations, for instance, tight focusing effects^[18] and Stokes vectors^[19–24] are used, the latter is also known as optical paraxial skyrmionic beams or Stokes skyrmions. Despite claims to the contrary, a Skyrmion has not yet been measured directly from a laser although it is likely produced in many prior lasers.^[25] Stokes skyrmions can be generated outside^[26,27] a laser cavity, and even in integrable on-chip devices,^[28] link the skyrmion number with orbital angular momentum (OAM), offering potential for optical communications,^[29–33] quantum entanglement source preparation^[34] and quantum storage.^[35–37] More importantly, OAM is theoretically infinite-dimensional in Hilbert space, making it a good candidate for generating higher-order skyrmions. Beyond the typical Laguerre-Gaussian (LG) beams, perfect vortex (PV) beams also carry OAM.^[38–41] The size of PV beams remains constant even as OAM increases, which is distinctly different

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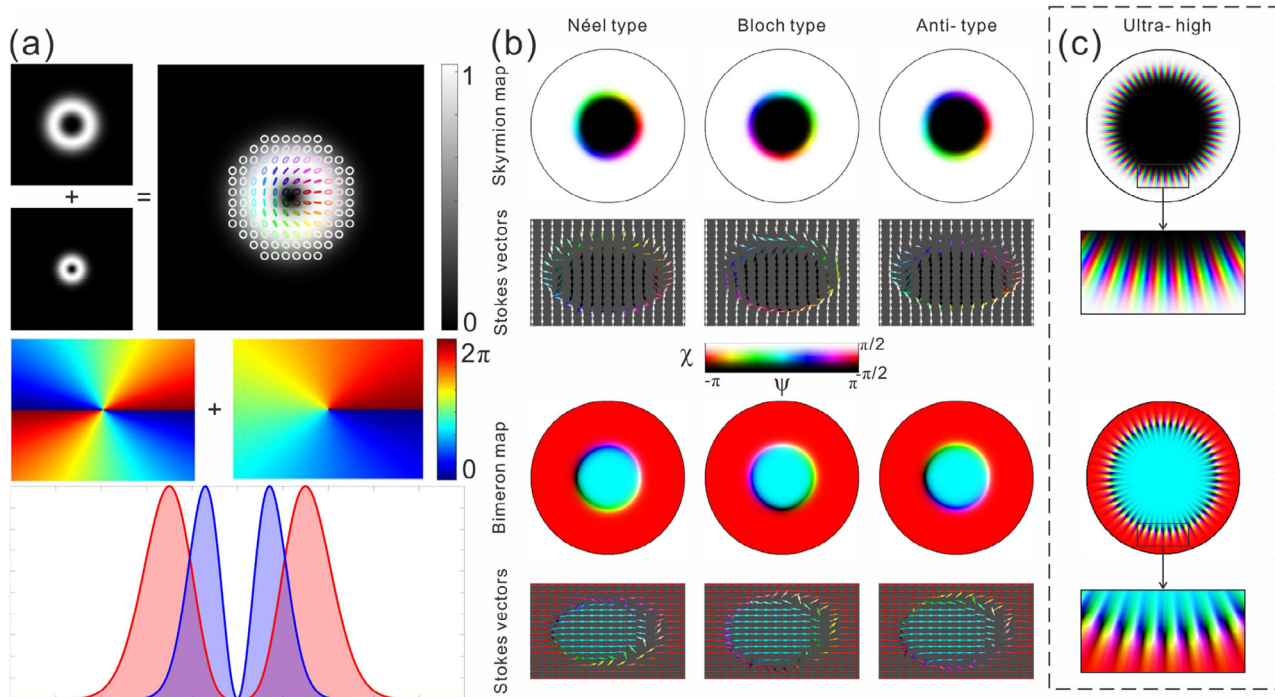


Figure 1. Theoretical concept of ring-shaped skyrmions and bimerons. a) show the amplitude and phase of two scalar vortex beams with orthogonal polarizations to vary spatially in such a way that the polarization of the superposed vector mode can cover the entire basic Poincaré sphere. The curve below shows the radial distribution of the intensity of the two scalar beams. b) are different ring-shaped types, with columns from left to right representing Néel, Bloch, and Anti-types. Top row, skyrmion maps and stokes vectors, the central black and peripheral white represent left- and right-handed circular polarization respectively; bottom row, the bimeron maps and stokes vectors, the central blue and peripheral red represent vertical and horizontal linear polarization respectively. Colorbar is based on the basic Poincaré sphere's azimuth angle Ψ and spatial angle χ . c) presents the maps of ultra-high-order skyrmions and bimerons along with their local magnified images. The main feature of the ring-shaped quasiparticles is that the center is no longer point-like but disk-like. The simulations are all superposed by LG beams with opposite unequal-order orbital angular momentum.

from LG modes. This makes them more favored in light-matter interaction applications^[42–45] such as particle manipulation^[46,47] and optical memory.^[48,49]

In this work, we investigate the feasibility of creating ring-shaped optical quasiparticles through topological approaches and demonstrate the experimental generation of ultra-high-order skyrmionic beams via the coherent superposition of typical or perfect optical modes. Such high-order skyrmions have never been generated in any physical systems. We not only generate the highest 400th-order skyrmions, significantly boosting information capacity, but also introduce a scheme for controlling the transverse beam size. The application of magnetic quasiparticles in high-density data storage^[50,51] gives us confidence to anticipate the future use of optical quasiparticles in optical memory and even the construction of topologically enhanced classical and quantum information networks.

2. Concepts

In mathematics, a 2D sphere in a real number space is topologically equivalent to the union of a 2D real number plane with the point at infinity:

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\} \cong \mathbb{R}^2 \cup \{\infty\} \quad (1)$$

this implies that there is always a continuous bijective mapping between these two topological spaces. Considering the South Pole

of the unit sphere tangent to the 2D real plane, the following stereographic projection can be constructed:

$$f : (x, y, 0) \mapsto \left(\frac{4x}{x^2 + y^2 + 4}, \frac{4y}{x^2 + y^2 + 4}, \frac{2(x^2 + y^2)}{x^2 + y^2 + 4} \right) \quad (2)$$

it is evident that f is invertible, and both f and f^{-1} are continuous mappings. As points on the plane $(x, y, 0)$ extend toward infinity, with $x^2 + y^2$ being an order of magnitude greater than x and y , the corresponding points on the sphere approach coordinates of 0 in the first two dimensions, while the z -coordinate $2 - 8/(x^2 + y^2 + 4)$ converges to 2, corresponding to the North Pole of the sphere. Another unique point is the tangency between the sphere and the 2D real plane at $O = (0, 0, 0)$, which corresponds to both the South Pole of the sphere and the origin of the plane. If the two topological spaces in Equation (1) are parameterized, for paraxial optics, such as the Stokes vectors, then the aforementioned mapping process corresponds precisely to the construction of a 2D skyrmionic structure ('baby skyrmions'^[17]), as shown in Figure 1.

Optical Skyrmions are typically described as coherent combinations of orthogonally polarized vortex and non-vortex beam, exemplified by LG beams:

$$E_s(r, \phi) = \cos(\theta) LG_0^{l_1}(r, \phi) \mathbf{e}_r + \sin(\theta) LG_0^{l_2}(r, \phi) \exp(i\Phi) \mathbf{e}_l \quad (3)$$

where $LG_0^l(r, \phi) = \sqrt{2/(\pi|l|!)}(\sqrt{2r/w_0})^{|l|} \exp(-r^2/w_0^2) \exp(il\phi)$ is the Laguerre-Gaussian function, w_0 is the beam waist, $|l_2| > l_1 = 0$ and $\theta \in [0, \pi/2]$ represents the relative amplitude of the two modes, Φ denotes the relative phase, and $\mathbf{e}_r, \mathbf{e}_\phi$ represent the unit vectors for right circularly polarized (RCP) and left circularly polarized (LCP) components, respectively. Such approach is effective for generating low-order skyrmionic beams, but it becomes quite different for higher-order cases. To illustrate this, we first need to understand that for the 3D Stokes vector $\mathbf{S}$, a topological invariant known as the winding number (or topological charge) can be defined as ref. [52]:

$$N = \frac{1}{4\pi} \iint \mathbf{S} \cdot \left(\frac{\partial \mathbf{S}}{\partial x} \times \frac{\partial \mathbf{S}}{\partial y} \right) dx dy \quad (4)$$

in polar and spherical coordinates, $\mathbf{r} = (r \cos \phi, r \sin \phi)$ and $\mathbf{S} = (\cos \Phi(\phi) \sin \theta(r), \sin \Phi(\phi) \sin \theta(r), \cos \theta(r))$, then Equation (4) is $\frac{1}{4\pi} \int_0^\infty dr \int_0^{2\pi} d\phi \frac{d\theta(r)}{dr} \frac{d\Phi(\phi)}{d\phi} \sin \theta(r) = \frac{1}{4\pi} \cos \theta(r)|_{r=0}^{r=\infty} \Phi(\phi)|_0^{2\pi} = m \cdot n$, where m, n is the polarity and vorticity. Clearly, the direction and times of the vectors covering the 3D parameter sphere determine the magnitude of N . Experimentally, to achieve a well-defined N and considering the spatial intensity distribution of the LG beam, the two modes in Equation (3) should have appropriate radial overlap. The spot size of LG beams is proportional to their order, hence higher-order ($|l| > 10$ here) LG beams will not have much effective overlap with the 0th-order mode, suggests an undesirable winding number.

To effectively generate higher-order skyrmions, l_1 in Equation (3) need not be 0 but should satisfy $|l_2| > |l_1|$. In this case, ‘baby skyrmions’ transform from disk-shaped to ring-shaped. We proceed with some intriguing topological manipulations to illustrate this idea: remove the South Pole of the parameter sphere and excise a circular disk from the 2D real plane centered at the origin. Define the remaining part of $\mathbb{R}^2$ as $X = \mathbb{R}^2 \setminus D$, where D is an open disk:

$$D = \{x \in \mathbb{R}^2 \mid \|x - O\| < r\} \quad (5)$$

where r is the radius of the disk. An equivalence relation can be defined on X : for $\forall(x, y) \in X$, if both x and y are on the boundary ∂D of D , then $x \sim y$; if only x is on ∂D , then $x \sim O$; otherwise, $x \sim x$. Thus, for the quotient space $Y = X / \sim$, all points on ∂D are equivalent to O , leading to the following quotient mapping:

$$p : X \rightarrow Y, x \mapsto \langle x \rangle \quad (6)$$

for $x \in \partial D$, $p(x) = O$; the rest $x \notin \partial D$ are mapped to their respective equivalence classes $\langle x \rangle$. Next, we need to define a surjection q : for $\forall x \in X$, if $x \notin \partial D$, then $q(\langle x \rangle) = x$; if $x \in \partial D$, then $q(\langle x \rangle) = O$. It is evident that this surjection is invertible and continuous, effectively showing that $Y \cong \mathbb{R}^2$. Through the mapping $p \circ q$, which compresses the boundary ∂D , we actually glue X and O into $\mathbb{R}^2$. In the Stokes parameter space, O represents a downward-pointing vector. The above mathematical analysis demonstrates that this vector’s point distribution forms a disk-like arrangement, thereby realizing the fundamental principle for constructing ring-shaped skyrmionic structures.

Figure 1a shows how to obtain a ring-shaped skyrmion by coherently superimposing scalar beams, simply by making two

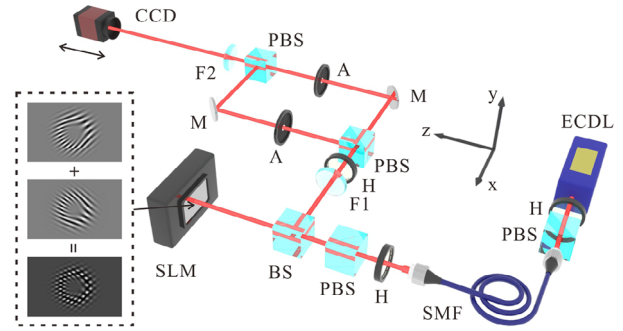


Figure 2. Experimental setup, ECDL: external cavity diode laser, H: half-wave plate, PBS: polarization beam splitter, SMF: single mode fiber, BS: beam splitter, SLM: spatial light modulator, F1, F2: lens with $f = 300$ mm, M: mirror, A: aperture, CCD: movable charge-coupled device, inset is the multiplexing grating. The Stokes parameters are measured using a polarizing analysis setup (not shown here) consisting of a HWP, a quarter-wave plate (QWP), and a PBS.

annular beams with orthogonal polarizations and different spot sizes form an effective superposition in space. It can be seen from the radial intensity distribution curve how to construct polarization that varies along the longitude of the basic Poincaré sphere. The blue curve is more dominant in the center while the red one is more prevalent at the edge. If the two represent LCP and RCP respectively, then the transition zone between them will exhibit linear polarization. On the other hand, since both of them carry vortex phases, the polarizations will evolve along the latitude of the basic Poincaré sphere. The top row of Figure 1b presents the map and Stokes vectors distributions for three types of ring-shaped 1st-order skyrmions. The central (boundary) vector directions of the three quasiparticles are uniformly downward (upward), with differences lying in the color transitions: the first column has a hedgehog-like structure, which is of the Néel type; the second column is defined as Bloch type, featuring a vortex structure; the last column is saddle-shaped, with the opposite topological charge compared to the first two, and is of the Anti-type. The bottom row of Figure 1 displays the ring-shaped bimeron structures, which are isomorphic to skyrmions and correspond to the three aforementioned types. It can be observed that each type of bimerons is composed of a pair of merons with the same $N = 1/2$ but opposite signs, similar to vortices pair before the topological phase transition occurs.^[53]

3. Experimental Section

The desired beam was generated by complex amplitude modulation of a spatial light modulator (SLM) with multiplexed gratings. As shown in Figure 2, an external cavity diode laser (ECDL) emits a horizontally polarized beam at 795 nm, which was coupled into a single-mode fiber (SMF) to improve the spatial mode. Due to the SLM’s ability to modulate only horizontally polarized light, the combination of a HWP and a PBS was employed prior to the beam’s incidence on the SLM. Additionally, a 50 : 50 beam splitter (BS) served to divide the unmodulated and modulated light.

Although the SLM modulated only the phase of the light, but a specific phase distribution could emulate complex amplitude

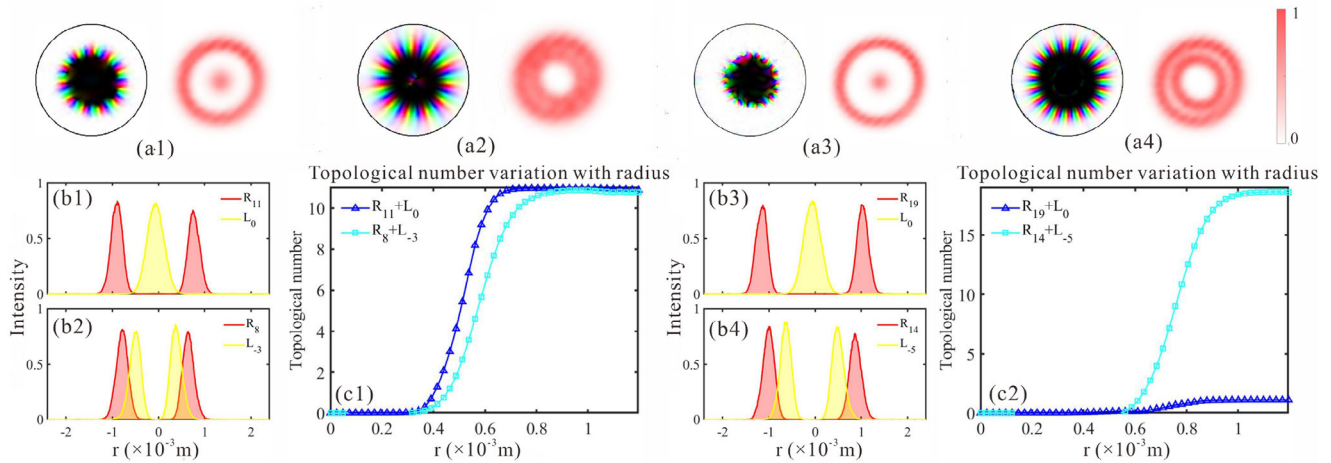


Figure 3. Comparison of experimental results of disk-shaped and ring-shaped high order skyrmions. The left is 11th-order skyrmions, (a1, a2), map and intensity distribution of superimposed modes: $R_{11} + L_0$ and $R_8 + L_{-3}$; (b1, b2), radial intensity curves of the two superimposed modes; (c1), topological number variation with integration radius ($\times 10^{-3}$ m). The right is 19th-order skyrmions, captions for the corresponding letters are the same as aforementioned, i.e., (a3, a4) for (a1, a2); (b3, b4) for (b1, b2); (c2) for (c1), while two superimposed modes are $R_{19} + L_0$ and $R_{14} + L_{-5}$. R_l and L_l is the l th RCP and LCP LG mode, respectively. The beam waist $w = 0.3$ mm.

modulation,^[54] as shown in the inset of Figure 2, where two orthogonally aligned blazed gratings produce a multiplexed grating through pointwise multiplication. To capture the complex amplitude-modulated beam from the SLM, a 4f imaging system was required between the SLM and the charge-coupled device (CCD). A Mach-Zehnder (MZ) interferometer was positioned between the two lenses. The modulated horizontally polarized beam first passes through a HWP oriented at $\pi/8$ to the horizontal, converting it into diagonally polarized light. The PBS then splits the modulated light into two beams of equal intensity, directing them to the two arms of the interferometer. Each arm had a spatial filter inserted to select the +1 order diffracted beams from each grating on the SLM. The output from the MZ interferometer represented the two orthogonally polarized modes described by Equation (3). By precisely adjusting the mirrors and PBSs, the two modes were coherently combined to generate the desired beam, which is then detected by a movable CCD.

4. Results and Discussions

4.1. Nonzero-Unequal Orders LG Beams as Basic Modes

To validate our approach, we present experimental results demonstrating the generation of ultra-high-order skyrmions. Figure 3 presents the generated distinct disk and ring-shaped skyrmions of 11th and 19th-order using LG modes. After applying Equation (4) and performing some noise reduction on the images, for typical method, the calculated winding number of the two high-order skyrmions are $N = 10.7636, 0.9975$, respectively. As shown in Figure 3a3, b3, the 19th-order modes overlap minimally with the 0th one in terms of intensity distribution, thus the skyrmion map transitioning chaotically from the dark center to the bright periphery. The red curve in

Figure 3c2, which shows the topological number variation with integration radius, confirms this observation. Although the 11th-order skyrmion maintains a robust topological structure, the 19th-order case significantly deviates from expectations. But for the ring-shaped case the calculated topological number are $N = 10.7636, 18.8301$, which confirmed the efficacy in generating higher-order skyrmions. However, when $|l|$ and r exceed critical values, the $LG_0^l(r, \phi)$ itself will lead to a clash: $r^{|l|} \rightarrow \infty$, $\exp(-r^2) \rightarrow 0$. Here, we prevent this by approximating with $LG_0^l(r, \phi) \approx \exp(-((r - w_0\sqrt{|l|/2})/w_0/\sqrt{2})^2) \exp(il\phi)$.

Figure 4 presents the results of generated ultra-high-order skyrmions. The calculated winding numbers are $N = 26.7343, 49.7268, 99.8110, 199.291, 296.5260, 392.0620$ for the 27th, 50th, 100th, 200th, 300th, and 400th-order skyrmions. From the magnified view of the skyrmion map, it is evident that the angular periodicity increases with the order, implying the Stokes vectors wrap around the parameter sphere along the latitude more times. As the topological charge increases, the azimuthal polarization exhibits more periodic variations, resulting in ultra-high-order skyrmions that primarily display moiré fringe patterns. From the curves of the topological numbers of the skyrmions of different orders versus integration shown in Figure 4b, it can be seen that the radial distance occupied by the polarization transition from the black center to the white edge is the same, and the slopes of the corresponding curves are not significantly different. This indicates that the method we used can ensure the effective overlap of the two orthogonal modes. Additionally, the overall shift of the curve to the right indicates that the size of the generated quasiparticles increases with the order. It should be noted that weak light intensities detected by the CCD in experiments can introduce noise, but proper data processing can mitigate its impact on the results (referring to color maps, with negligible effects on topological numbers), and during the polarization projection measurement, we fine-tuned the position of the PBS to achieve the optimal extinction ratio.

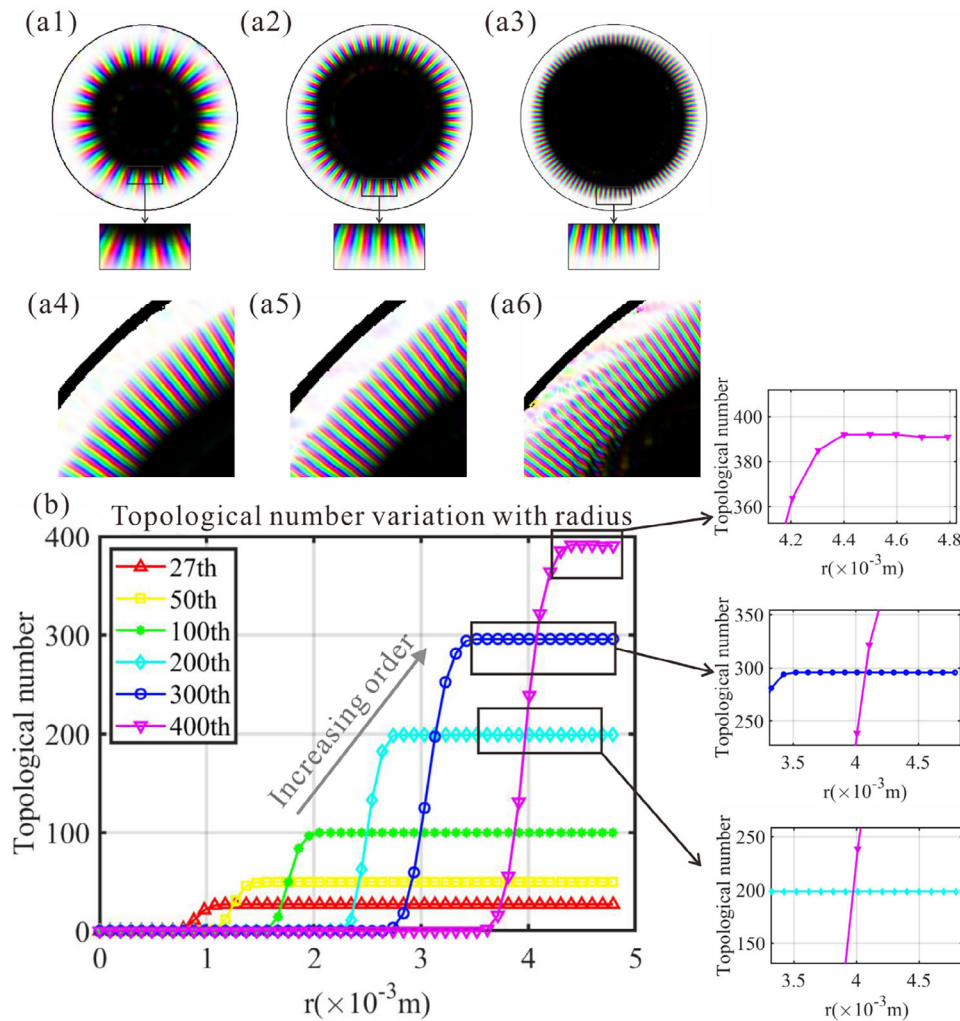


Figure 4. Experimental results of ultra-high-order skyrmions. a1–a3) is 27th, 50th, and 100th-order skyrmion with each local magnified image, respectively; due to the excessive number of overall polarization variation cycles, a4–a6) only show the magnified views of the 200th, 300th, and 400th-order skyrmions. b) is topological number variation with integration radius ($\times 10^{-3}$ m) of each ultra-high-order skyrmion with local magnified plots, the gray arrow indicates the trend of curve changes. The beam waist is the same as Figure 3.

Although we have confirmed that superimposing vortex beams of unequal order with orthogonal polarizations can yield good topological structures, there is still an upper limit to the order of skyrmions we can produce experimentally.^[55] Consider the following aspects. 1): Physically, high OAM density can lead to the generation of evanescent waves, hence for an optical system with a numerical aperture of NA, the topological number must satisfy:^[55]

$$|N| \leq k NA R \quad (7)$$

where k is the wave vector, R is the spot radius. 2): Experimentally, the screen size and resolution of the SLM and CCD are finite. Constrained by the characteristic that the LG mode's spot size increases with order, the SLM screen must be capable of modulating larger spots, and the CCD screen should also receive spots of corresponding size (diffraction must be taken into account). Moreover, due to the calculation of topological charge

based on pixel integration, higher order skyrmions do not have enough pixels for reliable results, so the precision of calculations is particularly crucial. However, existing literature^[56] on multiplying OAM by loading specific phases onto input beams through diffraction is expected to raise the upper limit of the quasiparticle order that can be generated.

Under the dual constraints of physical and device limitations mentioned above, Figure 4b shows the topological charge varies with the integration radius. It is clear that higher orders correspond to larger radial coordinates for reaching the maximum charge. Note that the maximum integration radius here is under 5 mm, due to our SLM resolution of 1900×1200 with a single pixel width of $8 \mu\text{m}$. Although such ultra-high-order topologies can greatly boost information capacity, the large spot is not favored in applications such as optical storage.^[48] While reducing the beam waist size may appear to enhance practical applicability, such an approach inherently restricts the method to digital devices like SLMs. Moreover, the necessity for repeated

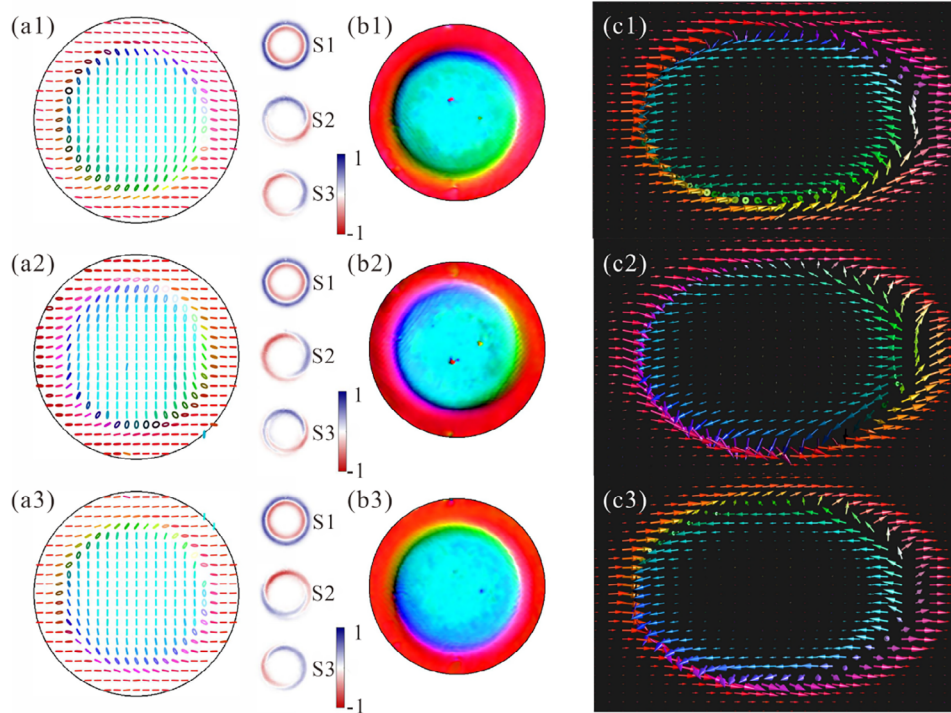


Figure 5. Experimental results of three different types 1st-order PB beams; The top row is Néel type, middle row is Bloch type, and bottom row is Anti-type; (a1–a3) are polarization distributions, insets are Stokes parameters; (b1–b3) are bimeronic maps; (c1–c3) are Stokes vectors distributions. Parameters used are: $T = 0.25$ mm, $R_1 = 5T$, $R_2 = 4T$.

parameter optimization ultimately compromises its practical utility.

4.2. PV Beam: Another Ring-Shaped Choice

An effective solution is to generate topological beams using PV beams. Typically, an ideal PV beam possesses arbitrary OAM and a fixed ring width, which can be described by a Dirac delta function, $\delta(r - R) \exp(il\phi)$, where R is the ring radius and l is OAM number. However, this cannot be experimentally realized, a quasi-PV beam with a finite thickness T is an excellent candidate.^[39,40]

$$P_{R,T}^l(r, \phi) = i^{l-1} \exp\left[-\frac{(r-R)^2}{T^2}\right] \exp(il\phi), R \gg T \quad (8)$$

by coherently combining two PV beams with distinct ring radius and OAM, a perfect topological structure is constructed (here we use perfect bimeronic (PB) beam as an example):

$$PB(r, \phi) = \cos(\theta) P_{R_1,T}^{l_1}(r, \phi) \mathbf{e}_x + \sin(\theta) P_{R_2,T}^{l_2}(r, \phi) \exp(i\Phi) \mathbf{e}_y, \quad (9)$$

where $l_1 \neq l_2$ & $R_1 \neq R_2$, θ and Φ are defined the same as Equation (3) mentioned above, and $\mathbf{e}_x$, $\mathbf{e}_y$ represent the unit vectors for horizontal and vertical polarizations, respectively.

Figure 5 presents the results of three types of bimerons generated in the experiment. It's clear that the experimental results align well with the simulated outcomes presented in Figure 1,

with minor shifts in the positions of LCP and RCP (dark and bright part in skyrmion map) due to the non-strict alignment of the two PV modes, yet it has minimal impact on the outcomes. The calculated topological charges of the three 1st-order bimerons are found to be: $N = 0.9923, 0.9893, -0.9923$. The experimental parameters such as R , T are provided in the caption of Figure 5.

Given that the key feature and advantage of PV beams is that their ring width does not increase with higher OAM values, the experiment successfully generated PB beams of higher orders, as shown in Figure 6a1–a5. First, the topological charges of the generated bimerons are calculated: $N = 1.9830, 2.9765, 4.9425, 9.8554, 19.6612$ for the 2nd, 3rd, 5th, 10th, and 20th-order. Figure 6b presents the variation of the topological number with integration radius. Intuitively, the integration values for PB beams of various orders transit from 0 to N between the center to the edge, indicating the Stokes vectors are well localized within the annular region. Moreover, the radial position of the max charge suggests that the PB beams of each order indeed have the same ring width, as more clearly shown in Figure 6c. The normalized radial intensity curves of PB beams of different orders essentially overlap. The irregularity of the curves makes it challenging to directly solve the ring width of PB beams, but we can use the difference in radial coordinates at half-peak widths of the two intensity peaks as the ring width. This is illustrated in the inset of Figure 6c, as the order increases, the ring width fluctuates by approximately just 0.01 mm.

Figure 6 d1-d2 show the experimental results of attempting to generate ultra-high-order perfect skyrmionic (PS) beams, param-

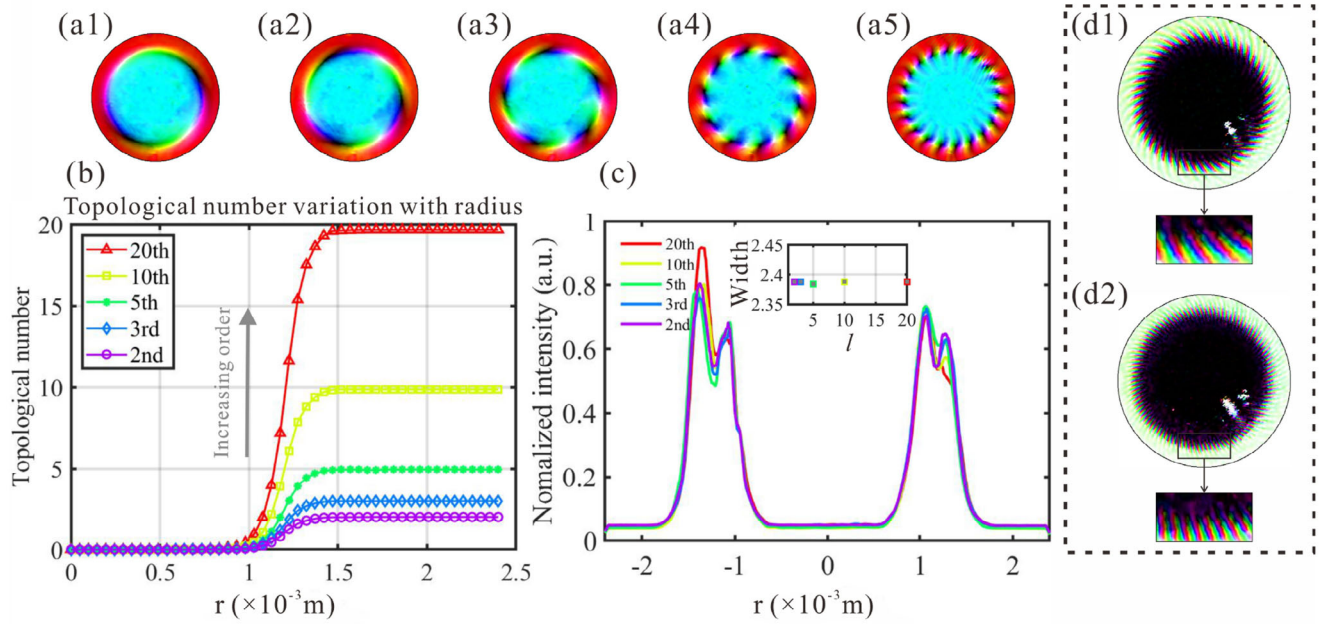


Figure 6. (a1–a5): Experimental results of 2nd, 3rd, 5th, 10th, 20th-order PB beams, parameters are the same as Figure 5; (b), topological number variation with integration radius ($\times 10^{-3}$ m), the gray arrow indicates the trend of curve changes; (c), radial normalized intensity, inset are ring widths of different orders; (d1,d2), results of 50th and 100th order perfect skyrmions and local magnified images, with $T = 0.1$ mm, $R_1 = 17T$, $R_2 = 14T$, $l_1 = 30, 60$, $l_2 = -20, -40$.

eters used are provided in the caption. The calculated winding numbers are $N = 49.6205, 96.0689$ of 50th and 100th-order PS beams, their ring widths are of course the same. Unlike the LG mode, the narrower thickness of PV beams on the SLM leads to weaker target field intensity and thus more noise. Besides, high-order Bessel-Gaussian (BG) beams as the Fourier transforms of PV beams have wide rings, making spatial filtering challenging. Existing literature^[40] has established the relationship between the asymptotic ring width of a single PV beam and OAM, for a PS beam ring width $w(l)$:

$$w(l) \propto (\sqrt{T^2(l+1) + 2R_1^2} + \sqrt{T^2l + 2R_2^2})/2 \quad (10)$$

where R_1, R_2 are ring radius of two PV beams in Equation (9) when $l = 0$, T is thickness. Equation (10) indicates that for larger l , R/T should be maximized to maintain a constant ring width, which will inevitably reduce the effective modulation intensity of the SLM. That said, the 100th-order PS beam with a smaller ring width we present here is already sufficient as an ultra-high capacity information carrier, and unlike the inherent limitations of LG modes, these drawbacks may be resolved through technological advances in the future. So far, we have proposed using LG modes and PV modes to construct ultra-high-order skyrmions, but which one is better? In fact, they each have their own strengths and limitations. Given the numerical aperture (NA) of the optical system and the same spot radius R , the upper limit of the topological number (N) that can be generated for LG and PV modes should be the same according to Equation (7). Since PV modes can carry higher-order OAM at a fixed R , they appear to be superior to LG modes. However, Equation (10) indi-

cates that the ability of PV modes to increase OAM while maintaining a constant spot radius is limited. On the other hand, similar to the azimuthal topological number resolution limit given by Equation (7), there should also be a limit to the topological number of the beam in the radial direction. Although there is no standard formula to describe this, we can imagine that when N is very high, due to the smaller effective overlapping area in the radial direction of the PV modes, the effective topological number will be significantly reduced. Thus, we say that PV modes are advantageous for constructing general high-order skyrmions, while LG modes should be preferred for constructing ultra-high-order skyrmions.

Finally, the free-space propagation characteristics of PB beams were investigated. Ideally, as an integer, the winding number remains unchanged under continuous deformations of the mapping (corresponding to propagation), thus serving as an invariant for the homotopy groups. To illustrate this, we generally need to construct a continuous mapping $H(s, z)$ to demonstrate the mappings at different z coordinates along the propagation direction are homotopic, that is:

$$C = \{h : S^2 \rightarrow Y \cup \{\infty\} \mid h \text{ is continuous}\} \quad (11)$$

for $\forall h_1, h_2 \in C$, there should be a continuous mapping $H : S^2 \times [0, 1] \rightarrow Y \cup \{\infty\}$ which satisfies:

$$H(s, 0) \equiv h_1(s), H(s, 1) \equiv h_2(s) \quad (12)$$

where Y has been defined above. Considering the complexity of constructing an explicit form of H mathematically, let us ap-

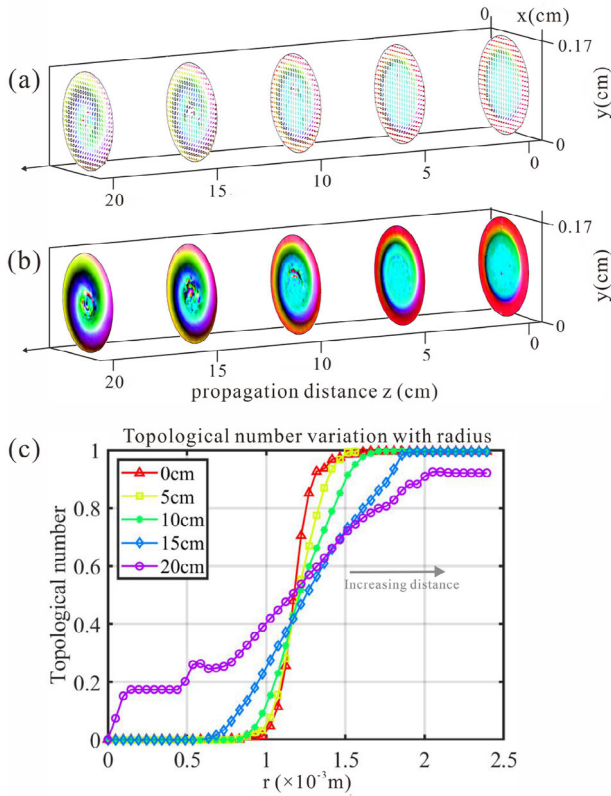


Figure 7. Propagation properties of 1st-order Néel-type PB beam (the top row of Figure 5) in z direction. a) changes of polarization during propagation; b) changes of bimeronic maps; c) topological number variation at different z , the gray arrow indicates the trend of curve changes. Data was collected at positions $z = 0, 5, 10, 15$, and 20 cm.

proach this issue from a more qualitative perspective.^[57] The complex amplitude of a single PV beam at z is ref. [39]:

$$P_R^l(r, \phi, z) = (-1)^l F(r, z) I_{l,R}(r, z) \cdot \exp[i\Phi_R(r, z)] \exp[i(\Psi + l\phi + kz)] \quad (13)$$

where $F(r, z) = \frac{T}{T_z} \exp[-\frac{1}{T_z}(r^2 - \frac{T_z^2}{z^2})]$, $T_z = T\sqrt{1 + (z/z_r)^2}$ is the thickness at z , $I_{l,R}(r, z) = I_l(\frac{2Rr\exp^{i\Psi}}{T_z})$, $I_l(\cdot)$ is the l th-order modified Bessel function of first kind, $\Phi_R(r, z) = \exp[\frac{ik}{2R_c}(r^2 + R^2)]$, $R_c = z + z_r^2/2$, $\Psi = \arctan(z/z_r)$ is the Gouy phase, $z_r = kT^2/2$ is the Rayleigh range. From Equation (13), one can see that for the two PV beams in Equation (9) with different l , R , the propagation results in only an overall scaling according to $I_{l,R}$ and a phase difference proportional to Φ_R . The polarization of PB beams merely rotates and twists continuously, which corresponds to the action of $H(s, z)$ on the mappings and thereby proves the topological stability during propagation.

Figure 7a,b respectively present the polarization and the bimeron map of a 1st-order Néel-type PB beam during propagation experimentally. Equation (13) shows that OAM only affects the phase term, hence it is predictable that higher-order cases will

be similar. Intuitively, the polarizations at each z indeed undergo continuous changes of rotation and overall twisting, which aligns with theoretical expectations. To more clearly verify the topological stability during propagation, Figure 7c depicts the relationship between the topological charge and the integration radius at different z values. Before propagating to $z = 20$ cm, all PB beams effectively achieve $N = 1$, but subsequent N values decrease due to excessive noise. The slope in Figure 7c diminishes with increasing distance. While PV patterns may fade over longer distances, a 20 cm propagation distance is adequate for most applications like quantum memory.^[58] For the PV mode with a thickness (T) of 0.25 mm used in the experiment, the Rayleigh distance (z_r) is approximately 25 cm, making 20 cm equivalent to $4/5z_r$. The PV mode evolves into the BG mode only after about $3z_r$, so the topological properties of the PB beam can be maintained over shorter distances. Enhanced noise reduction would boost measurement accuracy.

5. Conclusion

By employing the coherent superposition of diverse structured beams (including LG and PV modes), we demonstrate an effective approach for generating ultra-high-order optical skyrmions. Unlike conventional methods that create disk-shaped skyrmions through vortex/non-vortex beam combinations, our technique maps photonic spin textures onto ring-shaped domains while preserving their topological characteristics.

Our experiments demonstrate the generation of optical skyrmions with topological charges up to the 400th order using two LG beams with distinct, non-zero OAM indices - a significant advancement beyond previously reported limits. When employing PV modes, we observe that the resulting skyrmionic structures maintain a constant waist size regardless of their topological charge, providing an optimal solution for simultaneously achieving high-order configurations and compact beam profiles. Lastly, we experimentally confirmed the topological stability of ring-shaped optical skyrmions under continuous deformation, using the propagation characteristics of the PV mode as an example. The topologically protected ultra-high winding number, as a homotopy invariant, makes ring-shaped optical skyrmion a robust platform for high-capacity, disturbance-resistant classical, and quantum communications.

Appendix

To more clearly present the polarization distribution of the ultra-high-order Stokes skyrmions we generated, Figure A1 and Figure A2 show the Stokes parameters of the vector beams obtained by coherent superposition of LG and PV beams, respectively.

Since the Stokes parameter S_0 represents the complex amplitude distribution in the beam cross-section, it is evident why we refer to the generated ultra-high-order skyrmion as ring-shaped. The Stokes parameters S_1 and S_2 represent the distributions of horizontal and diagonal polarizations, respectively. For a skyrmion, they should show $2N$ alternating red and blue stripes along the azimuthal direction, meaning there are 800 stripes in Figure A1 and 200 stripes in Figure A2. Due to the limited number of pixels occupied by S_1 and S_2 within a single azimuthal period, only Moiré fringes are visible in the overall view. Therefore, local magnified images are provided. The results of the magnified images show that S_1 and S_2 differ by a $\pi/4$ rotation angle. Combined with precise numerical integra-

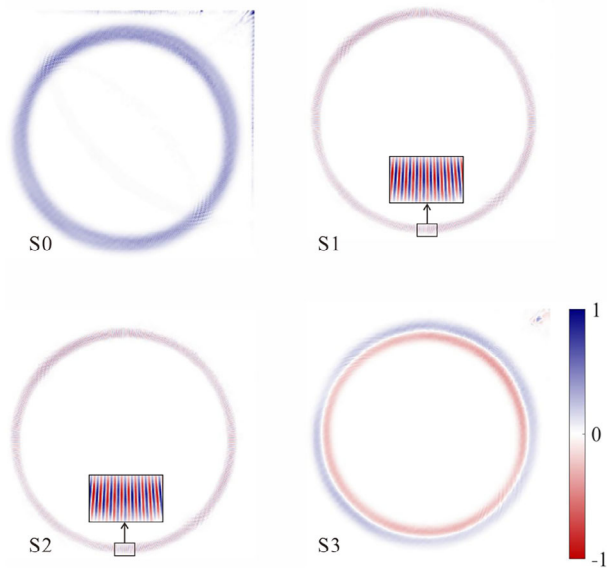


Figure A1. The experimental Stokes parameters $S_0 - S_3$ of the 400th-order skyrmion generated by LG beams, along with local magnified images of S_1 and S_2 .

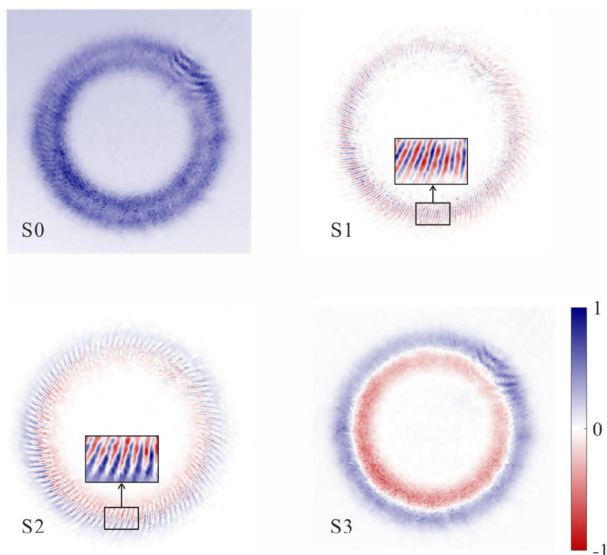


Figure A2. The experimental Stokes parameters $S_0 - S_3$ of the 100th-order skyrmion generated by PV beams, along with local magnified images of S_1 and S_2 .

tion methods, we obtained the better ultra-high-order topological numbers presented in the main text.

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Conflict of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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